Final Project - NFL Draft Career Trajectory Analysis

Benjamin Cotton
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Section 1 - Research Question

How does on-field production evolve across the first five seasons of a skill-position player's NFL career, and does draft round predict not just how much a player produces but how quickly they develop?

This question is interesting because draft position is often justified by teams as a signal of long term potential, not just immediate production. If first round picks produce more in year 1 but later round picks close the gap by year 3 or 4, that would suggest teams are overpaying for early production rather than true career value. I expected to find that first round picks dominate early but that the gap narrows significantly by season 3.

This question requires all three datasets to answer. draft_picks provides draft round and position group for each player. advstats provides season level rushing and receiving stats needed to track year by year production. combine provides measurables that let us ask whether physical traits explain any of the development gap between rounds. No single table alone can answer whether development trajectories differ by draft round.

This project builds on MA3. MA3 summarized career totals and single season averages grouped by draft round. This project goes further by introducing career year as a derived variable, computing each player's season number relative to their draft year, which allows us to compare how production changes over time rather than just how much players produce in aggregate.

Section 2 - Data Sources

Dataset	Source	Description	Scope	Collected by
`draft_picks.csv`	nflverse / PFR	Every NFL draft pick 1980–present. Includes round, pick number, position, college, team,	1980–present, all NFL teams	nflverse project aggregating Pro Football Reference data

Dataset	Source	Description	Scope	Collected by
		and career Approximate Value.		
`advstats_season_rush.csv` / `advstats_season_rec.csv`	nflverse / PFR	Advanced rushing and receiving stats per player per season. Includes yards, touchdowns, broken tackles, yards before/after contact.	2018–present, all NFL players	nflverse project aggregating Pro Football Reference data
`combine.csv`	nflverse / PFR	NFL Scouting Combine measurables including height, weight, 40-yard dash, bench press, vertical jump, and shuttle times.	2000–present, combine participants	nflverse project aggregating Pro Football Reference data

Limitations:

- `draft_picks`: Career AV (`w_av`) is empty in the current version of the file. This limits career level analysis to what is available in advstats.
- `advstats`: Only covers 2018–present, so career trajectory analysis is limited to players drafted 2018 or later. Players drafted before 2018 are excluded entirely.
- `combine`: Many players do not participate in the combine or have missing measurables, which limits how broadly combine analysis can be applied.

Section 3 - Load Libraries and Data

In [5]:
```
import sqlite3
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

pd.set_option('display.max_columns', 20)
pd.set_option('display.max_colwidth', 55)
```

In [6]:
```
# Load CSVs — same as MA3
draft = pd.read_csv('draft_picks.csv', low_memory=False)
combine = pd.read_csv('combine.csv', low_memory=False)
rush = pd.read_csv('advstats_season_rush.csv', low_memory=False)
rec = pd.read_csv('advstats_season_rec.csv', low_memory=False)
```

```
rush['stat_type'] = 'rush'
rec['stat_type'] = 'rec'
advstats = pd.concat([rush, rec], ignore_index=True)

print('draft_picks:', draft.shape)
print('combine:', combine.shape)
print('advstats:', advstats.shape)
```

```
draft_picks: (12670, 36)
combine: (8968, 18)
advstats: (6950, 30)
```

In [7]:
```
print('draft dtypes:')
print(draft.dtypes)
draft.head(3)
```

```
draft dtypes:
season                int64
round                 int64
pick                  int64
team                 object
gsis_id              object
pfr_player_id        object
cfb_player_id        object
pfr_player_name      object
hof                    bool
position             object
category             object
side                 object
college              object
age                 float64
to                  float64
allpro                int64
probowls              int64
seasons_started       int64
w_av                float64
car_av              float64
dr_av               float64
games               float64
pass_completions    float64
pass_attempts       float64
pass_yards          float64
pass_tds            float64
pass_ints           float64
rush_atts           float64
rush_yards          float64
rush_tds            float64
receptions          float64
rec_yards           float64
rec_tds             float64
def_solo_tackles    float64
def_ints            float64
def_sacks           float64
dtype: object
```

Out[7]:

	season	round	pick	team	gsis_id	pfr_player_id	cfb_player_id	pfr_player_nam
0	1980	1	1	DET	SIM659150	SimsBi00	billy-sims-1	Billy Sin
1	1980	1	2	NYJ	JON491656	JoneLa00	lam-jones-1	Lam Jone
2	1980	1	3	CIN	00-0011825	MunoAn00	NaN	Anthony Munc

3 rows × 36 columns

In [8]:
```
print('combine dtypes:')
print(combine.dtypes)
combine.head(3)
```

```
combine dtypes:
season           int64
draft_year     float64
draft_team      object
draft_round    float64
draft_ovr      float64
pfr_id          object
cfb_id          object
player_name     object
pos             object
school          object
ht              object
wt             float64
forty          float64
bench          float64
vertical       float64
broad_jump     float64
cone           float64
shuttle        float64
dtype: object
```

Out[8]:

	season	draft_year	draft_team	draft_round	draft_ovr	pfr_id	cfb_id	player_
0	2000	2000.0	New York Jets	1.0	13.0	AbraJo00	NaN	Ab
1	2000	2000.0	Seattle Seahawks	1.0	19.0	AlexSh00	shaun-alexander-1	Alex
2	2000	2000.0	Kansas City Chiefs	6.0	188.0	AlfoDa20	NaN	D

In [9]:
```
print('advstats dtypes:')
print(advstats.dtypes)
advstats.head(3)
```

```
advstats dtypes:
season            int64
player           object
pfr_id           object
tm               object
age               int64
pos              object
g                 int64
gs                int64
att             float64
yds               int64
td              float64
x1d             float64
ybc             float64
ybc_att         float64
yac             float64
yac_att         float64
brk_tkl         float64
att_br          float64
loaded           object
stat_type        object
tgt             float64
rec             float64
ybc_r           float64
yac_r           float64
adot            float64
rec_br          float64
drop            float64
drop_percent    float64
int             float64
rat             float64
dtype: object
```

Out[9]:

	season	player	pfr_id	tm	age	pos	g	gs	att	yds	...	tgt	rec
0	2018	Ezekiel Elliott	ElliEz00	DAL	23	RB	15	15	304.0	1434	...	NaN	NaN
1	2018	Saquon Barkley	BarkSa00	NYG	21	RB	16	16	261.0	1307	...	NaN	NaN
2	2018	David Johnson	JohnDa08	ARI	27	RB	16	16	258.0	940	...	NaN	NaN

3 rows × 30 columns

In [10]:
```
conn = sqlite3.connect('nfl_final.db')
cursor = conn.cursor()
```

Section 4 - Known Issues Log

Dataset	Column	Issue type	Description	Decision
draft_picks	pfr_player_id	Missingness	1,754 picks have no PFR ID, mostly very	Leave, these rows cannot join to

Dataset	Column	Issue type	Description	Decision
			old picks from before digital record keeping	advstats; documented as unmatched in join audit
draft_picks	position	Inconsistent category	PFR uses many position codes that map to the same group (OT, OG, T → OL)	Fix, add position_group with np.select
combine	wt	Invalid range	Weights below 150 or above 400 lbs are implausible	Fix, set to NaN
combine	forty	Missingness	Many players don't run the 40, especially linemen	Leave, expected; filter to skill positions when analyzing
advstats	pfr_id	Duplicates	One player appears once per season per stat_type — this is expected grain, not a true duplicate	Leave, by design

In [12]:
```
# Missingness  draft_picks
print('draft null counts:')
print(draft[['pfr_player_id','position','college']].isna().sum())
```

```
draft null counts:
pfr_player_id    1754
position            0
college            12
dtype: int64
```

In [13]:
```
# Missingness  combine
print('combine null counts:')
print(combine[['pfr_id','forty','wt']].isna().sum())
```

```
combine null counts:
pfr_id    1531
forty      929
wt          24
dtype: int64
```

In [14]:
```
# Inconsistent categories  position codes
print('Distinct position values in draft_picks:')
print(draft['position'].value_counts())
```

```
Distinct position values in draft_picks:
position
DB     1929
WR     1567
LB     1546
RB     1251
DE     1053
T       838
DT      809
TE      732
G       707
QB      597
C       313
CB      262
OL      225
S       146
K       123
P       119
NT      114
FB       86
OLB      85
DL       78
ILB      34
SAF      30
OT       13
LS        9
KR        2
FS        2
Name: count, dtype: int64
```

In [15]:
```
# Invalid range  weight
print('Weight range in combine:')
print(combine['wt'].describe())
print('Rows outside 150-400 lbs:', ((combine['wt'] < 150) | (combine['wt'] >
```

```
Weight range in combine:
count    8944.000000
mean      242.641324
std        45.701624
min       144.000000
25%       205.000000
50%       232.000000
75%       281.000000
max       384.000000
Name: wt, dtype: float64
Rows outside 150-400 lbs: 2
```

In [16]:
```
# Duplicates  advstats grain check
# Each player should appear once per season per stat_type
dupes = advstats.groupby(['pfr_id','season','stat_type']).size().reset_index
print('advstats rows with count > 1 (expect 0):')
print(dupes[dupes['cnt'] > 1].shape[0])
```

```
advstats rows with count > 1 (expect 0):
36
```

Section 5 - Data Dictionary

draft_picks

- `season` (INTEGER): NFL Draft year
- `round` (INTEGER): Draft round (1–7)
- `pick` (INTEGER): Overall pick number
- `team` (TEXT): Team abbreviation that made the pick
- `pfr_player_id` (TEXT): Unique player ID from Pro Football Reference, join key to combine and advstats
- `pfr_player_name` (TEXT): Player name
- `position` (TEXT): Raw position code from PFR
- `position_group` (TEXT): Standardized position group derived from position, QB, RB, WR, TE, OL, DL, LB, DB, ST, OTHER
- `college` (TEXT): College attended before being drafted
- `probowls` (INTEGER): Career Pro Bowl selections
- `seasons_started` (INTEGER): Seasons as a primary starter
- `w_av` (REAL): Weighted Career Approximate Value

advstats

- `pfr_id` (TEXT): Pro Football Reference player ID, join key
- `player` (TEXT): Player name
- `season` (INTEGER): NFL season year
- `tm` (TEXT): Team abbreviation that season
- `age` (REAL): Player age that season
- `pos` (TEXT): Position that season
- `g` / `gs` (INTEGER): Games played / started
- `stat_type` (TEXT): 'rush' or 'rec'
- `att` (INTEGER): Attempts (carries for rush, targets for rec)
- `yds` (REAL): Yards gained
- `td` (INTEGER): Touchdowns
- `ybc` / `yac` (REAL): Yards before/after contact or catch
- `brk_tkl` (INTEGER): Broken tackles
- `career_year` (INTEGER): Derived variable, number of seasons since the player was drafted (1 = draft year, 2 = second season, etc.)

combine

- `season` (INTEGER): Year of the NFL Scouting Combine
- `pfr_id` (TEXT): Pro Football Reference player ID, join key
- `player_name` (TEXT): Player name
- `pos` (TEXT): Position at the combine

- `school` (TEXT): College attended
- `ht` (REAL): Height in inches
- `wt` (REAL): Weight in pounds
- `forty` (REAL): 40-yard dash time in seconds
- `vertical` (REAL): Vertical jump in inches

Section 6 - Cleaning and Validation

In [19]:
```
# Fix 1  strip whitespace from text columns
for col in draft.select_dtypes(include='object').columns:
    draft[col] = draft[col].str.strip()
for col in combine.select_dtypes(include='object').columns:
    combine[col] = combine[col].str.strip()
```

In [20]:
```
# Fix 2  set implausible weight values to NaN
before = combine['wt'].isna().sum()
combine.loc[(combine['wt'] < 150) | (combine['wt'] > 400), 'wt'] = np.nan
after = combine['wt'].isna().sum()
print(f'wt nulls before: {before}  after: {after}  newly nulled: {after - be
```

```
wt nulls before: 24  after: 26  newly nulled: 2
```

In [21]:
```
# Fix 3  add position_group to standardize position codes
conditions = [
    draft['position'].isin(['QB']),
    draft['position'].isin(['RB','HB','FB']),
    draft['position'].isin(['WR']),
    draft['position'].isin(['TE']),
    draft['position'].isin(['OT','OG','OL','C','G','T']),
    draft['position'].isin(['DE','DT','NT','DL']),
    draft['position'].isin(['LB','ILB','OLB','MLB']),
    draft['position'].isin(['CB','DB','S','SS','FS','SAF']),
    draft['position'].isin(['K','P','LS'])
]
choices = ['QB','RB','WR','TE','OL','DL','LB','DB','ST']
draft['position_group'] = np.select(conditions, choices, default='OTHER')
print('position_group counts:')
print(draft['position_group'].value_counts())
```

```
position_group counts:
position_group
DB       2369
OL       2096
DL       2054
LB       1665
WR       1567
RB       1337
TE        732
QB        597
ST        251
OTHER       2
Name: count, dtype: int64
```

In [22]:
```
# Fix 4  compute career_year in advstats
# Join draft season onto advstats so we can compute stat_season - draft_seas
# career_year = 1 means the player's first NFL season (their draft year)
draft_year_lookup = draft[['pfr_player_id','season']].rename(
    columns={'pfr_player_id':'pfr_id', 'season':'draft_season'}
)
advstats = advstats.merge(draft_year_lookup, on='pfr_id', how='left')
advstats['career_year'] = advstats['season'] - advstats['draft_season'] + 1

# Keep only career years 1-5 and drop incomplete data
advstats = advstats[(advstats['career_year'] >= 1) & (advstats['career_year'
print('advstats rows after filtering to career years 1-5:', len(advstats))
print('career_year distribution:')
print(advstats['career_year'].value_counts().sort_index())
```

```
advstats rows after filtering to career years 1-5: 3373
career_year distribution:
career_year
1.0    812
2.0    777
3.0    684
4.0    605
5.0    495
Name: count, dtype: int64
```

In [23]:
```
# Load cleaned data into SQLite
draft_cols = [
    'season','round','pick','team','pfr_player_id','pfr_player_name',
    'position','position_group','category','side','college',
    'allpro','probowls','seasons_started','w_av','dr_av','games',
    'rush_yards','rush_tds','rec_yards','rec_tds','def_sacks'
]
combine_cols = [
    'season','pfr_id','player_name','pos','school',
    'ht','wt','forty','bench','vertical','broad_jump','cone','shuttle'
]
advstats_cols = [
    'pfr_id','player','season','draft_season','career_year',
    'tm','age','pos','g','gs','stat_type','att','yds','td','ybc','yac','brk_
]

draft_cols_use    = [c for c in draft_cols    if c in draft.columns]
combine_cols_use  = [c for c in combine_cols  if c in combine.columns]
advstats_cols_use = [c for c in advstats_cols if c in advstats.columns]

cursor.execute('DROP TABLE IF EXISTS draft_picks;')
cursor.execute('DROP TABLE IF EXISTS combine;')
cursor.execute('DROP TABLE IF EXISTS advstats;')
conn.commit()

draft[draft_cols_use].to_sql('draft_picks', conn, if_exists='replace', index
combine[combine_cols_use].to_sql('combine', conn, if_exists='replace', index
advstats[advstats_cols_use].to_sql('advstats', conn, if_exists='replace', ir
conn.commit()
```

In [24]:
```
# Validation  row counts
for table in ['draft_picks','combine','advstats']:
    n = pd.read_sql_query(f'SELECT COUNT(*) AS n FROM {table};', conn).iloc[
    print(f'{table}: {n} rows')
```

```
draft_picks: 12670 rows
combine: 8968 rows
advstats: 3373 rows
```

In [25]:
```
# Pre-join audit  cardinality
print('draft_picks: total rows vs distinct pfr_player_ids:')
display(pd.read_sql_query("""
SELECT COUNT(*) AS total_rows, COUNT(DISTINCT pfr_player_id) AS distinct_ids
FROM draft_picks WHERE pfr_player_id IS NOT NULL;
""", conn))

print('advstats: total rows vs distinct pfr_ids:')
display(pd.read_sql_query("""
SELECT COUNT(*) AS total_rows, COUNT(DISTINCT pfr_id) AS distinct_ids
FROM advstats WHERE pfr_id IS NOT NULL;
""", conn))
```

```
draft_picks: total rows vs distinct pfr_player_ids:
```

	total_rows	distinct_ids
0	10916	10914

```
advstats: total rows vs distinct pfr_ids:
```

	total_rows	distinct_ids
0	3373	849

In [26]:
```
# Pre-join audit  unmatched records
print('draft_picks pfr_player_ids with no match in advstats:')
display(pd.read_sql_query("""
SELECT COUNT(*) AS unmatched
FROM draft_picks
WHERE pfr_player_id IS NOT NULL
  AND pfr_player_id NOT IN (
      SELECT DISTINCT pfr_id FROM advstats WHERE pfr_id IS NOT NULL
  );
""", conn))

print('advstats pfr_ids with no match in draft_picks:')
display(pd.read_sql_query("""
SELECT COUNT(*) AS unmatched
FROM advstats
WHERE pfr_id IS NOT NULL
  AND pfr_id NOT IN (
      SELECT DISTINCT pfr_player_id FROM draft_picks WHERE pfr_player_id IS
  );
""", conn))
```

```
draft_picks pfr_player_ids with no match in advstats:
```

	unmatched
0	10067

advstats pfr_ids with no match in draft_picks:

	unmatched
0	0

In [27]:
```
# Pre-join audit  row count expectation
# draft_picks LEFT JOIN advstats is 1-to-many (one player, many season rows)
# We expect more rows than draft_picks alone
print('advstats rows for skill players drafted 2018+:')
display(pd.read_sql_query("""
SELECT COUNT(*) AS expected_joined_rows
FROM advstats
WHERE pfr_id IN (
    SELECT pfr_player_id FROM draft_picks
    WHERE season >= 2018
      AND position_group IN ('RB','WR','TE')
);
""", conn))
```

advstats rows for skill players drafted 2018+:

	expected_joined_rows
0	2331

In [28]:
```
# Final join
joined = pd.read_sql_query("""
SELECT
    d.season        AS draft_season,
    d.round,
    d.pick,
    d.pfr_player_name AS player_name,
    d.pfr_player_id,
    d.position_group,
    d.college,
    c.forty,
    c.wt            AS weight,
    c.vertical,
    a.season        AS stat_season,
    a.career_year,
    a.stat_type,
    a.yds,
    a.td,
    a.att,
    a.brk_tkl
FROM draft_picks d
LEFT JOIN combine  c ON d.pfr_player_id = c.pfr_id
LEFT JOIN advstats a ON d.pfr_player_id = a.pfr_id
WHERE d.season >= 2018
  AND d.position_group IN ('RB','WR','TE')
ORDER BY d.season, d.pick, a.career_year
```

```
LIMIT 200;
""", conn)

print('Join sample (first 20 rows):')
display(joined.head(20))
```

```
Join sample (first 20 rows):
```

	draft_season	round	pick	player_name	pfr_player_id	position_group	college	fort
0	2018	1	2	Saquon Barkley	BarkSa00	RB	Penn St.	4.4
1	2018	1	2	Saquon Barkley	BarkSa00	RB	Penn St.	4.4
2	2018	1	2	Saquon Barkley	BarkSa00	RB	Penn St.	4.4
3	2018	1	2	Saquon Barkley	BarkSa00	RB	Penn St.	4.4
4	2018	1	2	Saquon Barkley	BarkSa00	RB	Penn St.	4.4
5	2018	1	2	Saquon Barkley	BarkSa00	RB	Penn St.	4.4
6	2018	1	2	Saquon Barkley	BarkSa00	RB	Penn St.	4.4
7	2018	1	2	Saquon Barkley	BarkSa00	RB	Penn St.	4.4
8	2018	1	2	Saquon Barkley	BarkSa00	RB	Penn St.	4.4
9	2018	1	2	Saquon Barkley	BarkSa00	RB	Penn St.	4.4
10	2018	1	24	D.J. Moore	MoorD.00	WR	Maryland	4.4
11	2018	1	24	D.J. Moore	MoorD.00	WR	Maryland	4.4
12	2018	1	24	D.J. Moore	MoorD.00	WR	Maryland	4.4
13	2018	1	24	D.J. Moore	MoorD.00	WR	Maryland	4.4
14	2018	1	24	D.J. Moore	MoorD.00	WR	Maryland	4.4
15	2018	1	24	D.J. Moore	MoorD.00	WR	Maryland	4.4
16	2018	1	24	D.J. Moore	MoorD.00	WR	Maryland	4.4
17	2018	1	24	D.J. Moore	MoorD.00	WR	Maryland	4.4
18	2018	1	24	D.J. Moore	MoorD.00	WR	Maryland	4.4
19	2018	1	24	D.J. Moore	MoorD.00	WR	Maryland	4.4

In [29]:
```
# Post-join validation  row count
print('Full join row count:')
```

```
display(pd.read_sql_query("""
SELECT COUNT(*) AS total_rows
FROM draft_picks d
LEFT JOIN combine  c ON d.pfr_player_id = c.pfr_id
LEFT JOIN advstats a ON d.pfr_player_id = a.pfr_id
WHERE d.season >= 2018
  AND d.position_group IN ('RB','WR','TE');
""", conn))
```

```
Full join row count:
```

	total_rows
0	2373

Join quality report

I used LEFT JOIN for both combine and advstats to preserve all draft picks as the base unit of analysis, even when combine or career stats data is absent. The combine join is 1-to-1 on pfr_player_id = pfr_id and does not expand row count. The advstats join is 1-to-many, one player has one row per career year per stat type, so rows expand as expected and documented in the pre-join audit. Unmatched combine records are players who skipped the combine or have no recorded ID, they retain NULL combine values. Unmatched advstats records are drafted players who never accumulated rushing or receiving stats at the skill positions, which is expected.

Section 7 - Analysis

Production by career year, does output grow over time?

In [33]:
```
# Groups all skill position players by career year and stat type
# No minimum att filter, all seasons included
print('Average yards by career year (all rounds, skill positions):')
by_year = pd.read_sql_query("""
SELECT
    a.career_year,
    a.stat_type,
    COUNT(DISTINCT a.pfr_id) AS players,
    ROUND(AVG(a.yds), 1)     AS avg_yds,
    ROUND(AVG(a.td), 2)      AS avg_td,
    ROUND(AVG(a.brk_tkl), 2) AS avg_brk_tkl
FROM advstats a
JOIN draft_picks d ON a.pfr_id = d.pfr_player_id
WHERE d.position_group IN ('RB','WR','TE')
GROUP BY a.career_year, a.stat_type
ORDER BY a.career_year, a.stat_type;
""", conn)
display(by_year)
```

```
Average yards by career year (all rounds, skill positions):
```

	career_year	stat_type	players	avg_yds	avg_td	avg_brk_tkl
0	1.0	rec	442	258.5	1.61	1.62
1	1.0	rush	278	194.0	1.32	3.27
2	2.0	rec	427	309.4	1.82	2.03
3	2.0	rush	255	242.7	1.81	4.01
4	3.0	rec	371	343.7	1.99	2.31
5	3.0	rush	232	250.3	1.78	4.31
6	4.0	rec	342	314.0	1.78	1.78
7	4.0	rush	196	258.7	2.15	4.00
8	5.0	rec	274	342.4	2.15	1.82
9	5.0	rush	153	251.0	1.94	3.98

Development trajectories by draft round, do later-round picks close the gap?

In [35]:
```
print(pd.read_sql_query("""
SELECT career_year, COUNT(*) AS n
FROM advstats
GROUP BY career_year
ORDER BY career_year;
""", conn))
```

```
   career_year    n
0          1.0  812
1          2.0  777
2          3.0  684
3          4.0  605
4          5.0  495
```

In [36]:
```
print(pd.read_sql_query("""
SELECT d.round, a.career_year, COUNT(DISTINCT a.pfr_id) AS players
FROM advstats a
JOIN draft_picks d ON a.pfr_id = d.pfr_player_id
WHERE d.position_group IN ('RB','WR','TE')
  AND a.stat_type = 'rec'
  AND d.round <= 4
GROUP BY d.round, a.career_year
ORDER BY d.round, a.career_year;
""", conn))
```

	round	career_year	players
0	1	1.0	54
1	1	2.0	55
2	1	3.0	50
3	1	4.0	50
4	1	5.0	46
5	2	1.0	76
6	2	2.0	73
7	2	3.0	72
8	2	4.0	63
9	2	5.0	57
10	3	1.0	73
11	3	2.0	67
12	3	3.0	62
13	3	4.0	57
14	3	5.0	49
15	4	1.0	84
16	4	2.0	79
17	4	3.0	63
18	4	4.0	64
19	4	5.0	46

In [37]:
```
# Average receiving yards per career year broken down by draft round
# This is the core analysis, comparing year by year production across rounds
print('Average receiving yards by career year and draft round:')
traj_rec = pd.read_sql_query("""
SELECT
    d.round,
    a.career_year,
    COUNT(DISTINCT a.pfr_id) AS players,
    ROUND(AVG(a.yds), 1)     AS avg_rec_yds,
    ROUND(AVG(a.td), 2)      AS avg_td
FROM advstats a
JOIN draft_picks d ON a.pfr_id = d.pfr_player_id
WHERE d.position_group IN ('RB','WR','TE')
  AND a.stat_type = 'rec'
  AND d.round <= 4
GROUP BY d.round, a.career_year
HAVING COUNT(DISTINCT a.pfr_id) >= 5
ORDER BY d.round, a.career_year;
""", conn)
display(traj_rec)
```

Average receiving yards by career year and draft round:

	round	career_year	players	avg_rec_yds	avg_td
0	1	1.0	54	650.7	3.50
1	1	2.0	55	615.7	3.77
2	1	3.0	50	672.4	3.22
3	1	4.0	50	548.1	2.78
4	1	5.0	46	606.0	3.79
5	2	1.0	76	374.6	2.63
6	2	2.0	73	450.8	2.61
7	2	3.0	72	417.7	2.49
8	2	4.0	63	459.5	2.57
9	2	5.0	57	364.1	2.30
10	3	1.0	73	212.6	1.26
11	3	2.0	67	335.3	2.48
12	3	3.0	62	357.1	2.61
13	3	4.0	57	326.9	2.27
14	3	5.0	49	360.5	2.51
15	4	1.0	84	177.0	1.09
16	4	2.0	79	192.9	1.17
17	4	3.0	63	245.5	1.50
18	4	4.0	64	223.7	1.07
19	4	5.0	46	221.1	1.26

In [38]:
```
# Average rushing yards by career year and draft round
print('Average rushing yards by career year and draft round:')
traj_rush = pd.read_sql_query("""
SELECT
    d.round,
    a.career_year,
    COUNT(DISTINCT a.pfr_id) AS players,
    ROUND(AVG(a.yds), 1)     AS avg_rush_yds,
    ROUND(AVG(a.td), 2)      AS avg_td
FROM advstats a
JOIN draft_picks d ON a.pfr_id = d.pfr_player_id
WHERE d.position_group IN ('RB','WR','TE')
  AND a.stat_type = 'rush'
  AND d.round <= 4
GROUP BY d.round, a.career_year
HAVING COUNT(DISTINCT a.pfr_id) >= 5
ORDER BY d.round, a.career_year;
```

```
""", conn)
display(traj_rush)
```

Average rushing yards by career year and draft round:

	round	career_year	players	avg_rush_yds	avg_td
0	1	1.0	42	243.3	2.00
1	1	2.0	41	272.5	1.97
2	1	3.0	39	280.4	2.11
3	1	4.0	29	347.4	3.71
4	1	5.0	31	253.8	2.39
5	2	1.0	53	238.5	1.86
6	2	2.0	48	281.2	1.95
7	2	3.0	44	339.5	2.73
8	2	4.0	38	380.0	3.00
9	2	5.0	32	385.4	2.52
10	3	1.0	42	202.6	1.50
11	3	2.0	44	308.1	2.67
12	3	3.0	39	268.7	2.04
13	3	4.0	35	315.4	2.69
14	3	5.0	29	304.7	3.09
15	4	1.0	51	218.7	1.26
16	4	2.0	48	189.6	1.37
17	4	3.0	35	218.8	1.35
18	4	4.0	36	194.3	1.25
19	4	5.0	27	198.6	1.30

Year over year change, derived variable

In [40]:

```
# Derived variable: yds_change = avg yards in career year N minus avg yards
# yds_change is computed by merging the table against a shifted copy of itse
# the previous year's avg_rec_yds from the current year's.
# Career year 1 shows NaN since there is no prior year to compare against.
# Computed by merging the table with itself on round, offsetting career_year

yoy = pd.read_sql_query("""
SELECT
    d.round,
    a.career_year,
    ROUND(AVG(a.yds), 1) AS avg_rec_yds
FROM advstats a
```

```
JOIN draft_picks d ON a.pfr_id = d.pfr_player_id
WHERE d.position_group IN ('RB','WR','TE')
  AND a.stat_type = 'rec'
  AND d.round <= 4
GROUP BY d.round, a.career_year
HAVING COUNT(DISTINCT a.pfr_id) >= 5
ORDER BY d.round, a.career_year;
""", conn)

# Create a copy with career_year shifted by 1 so we can merge current year a
yoy_prev = yoy.copy()
yoy_prev['career_year'] = yoy_prev['career_year'] + 1
yoy_prev = yoy_prev.rename(columns={'avg_rec_yds': 'prev_yds'})

# Merge current year with previous year on round and career_year
yoy_change = yoy.merge(yoy_prev[['round','career_year','prev_yds']], on=['ro
yoy_change['yds_change'] = (yoy_change['avg_rec_yds'] - yoy_change['prev_yds

print('Year-over-year change in avg receiving yards by round:')
display(yoy_change[['round','career_year','avg_rec_yds','yds_change']])
```

```
Year-over-year change in avg receiving yards by round:
```

	round	career_year	avg_rec_yds	yds_change
0	1	1.0	650.7	NaN
1	1	2.0	615.7	-35.0
2	1	3.0	672.4	56.7
3	1	4.0	548.1	-124.3
4	1	5.0	606.0	57.9
5	2	1.0	374.6	NaN
6	2	2.0	450.8	76.2
7	2	3.0	417.7	-33.1
8	2	4.0	459.5	41.8
9	2	5.0	364.1	-95.4
10	3	1.0	212.6	NaN
11	3	2.0	335.3	122.7
12	3	3.0	357.1	21.8
13	3	4.0	326.9	-30.2
14	3	5.0	360.5	33.6
15	4	1.0	177.0	NaN
16	4	2.0	192.9	15.9
17	4	3.0	245.5	52.6
18	4	4.0	223.7	-21.8
19	4	5.0	221.1	-2.6

Comparison of round 1 vs round 3+, how big is the gap and does it close?

In [42]:
```
# CASE WHEN to bin picks into Early (rounds 1-2) vs Late (rounds 3-7)
# Then compare avg receiving yards by career year between the two tiers
print('Early (rounds 1-2) vs Late (rounds 3-7) receiving yards by career yea
tier_compare = pd.read_sql_query("""
SELECT
    CASE
        WHEN d.round <= 2 THEN 'Early (rounds 1-2)'
        ELSE                   'Late (rounds 3-7)'
    END AS draft_tier,
    a.career_year,
    COUNT(DISTINCT a.pfr_id) AS players,
    ROUND(AVG(a.yds), 1)     AS avg_rec_yds
FROM advstats a
JOIN draft_picks d ON a.pfr_id = d.pfr_player_id
```

```
WHERE d.position_group IN ('RB','WR','TE')
  AND a.stat_type = 'rec'
GROUP BY draft_tier, a.career_year
HAVING COUNT(DISTINCT a.pfr_id) >= 5
ORDER BY a.career_year, draft_tier;
""", conn)
display(tier_compare)
```

```
Early (rounds 1-2) vs Late (rounds 3-7) receiving yards by career year:
```

	draft_tier	career_year	players	avg_rec_yds
0	Early (rounds 1-2)	1.0	130	489.3
1	Late (rounds 3-7)	1.0	312	162.4
2	Early (rounds 1-2)	2.0	128	519.5
3	Late (rounds 3-7)	2.0	299	216.6
4	Early (rounds 1-2)	3.0	122	522.1
5	Late (rounds 3-7)	3.0	249	256.3
6	Early (rounds 1-2)	4.0	113	498.1
7	Late (rounds 3-7)	4.0	229	222.4
8	Early (rounds 1-2)	5.0	103	470.1
9	Late (rounds 3-7)	5.0	171	263.9

Does combine speed predict faster development?

In [44]:
```
# Split players into fast (forty < 4.5) and slow (forty >= 4.5) at skill pos
# Compare their avg receiving yards in career year 1 vs career year 3
print('Receiving yards by career year and combine speed tier:')
speed_traj = pd.read_sql_query("""
SELECT
    CASE
        WHEN c.forty < 4.5  THEN 'Fast (< 4.50s)'
        WHEN c.forty >= 4.5 THEN 'Slower (4.50s+)'
        ELSE                     'No forty recorded'
    END AS speed_tier,
    a.career_year,
    COUNT(DISTINCT a.pfr_id) AS players,
    ROUND(AVG(a.yds), 1)     AS avg_rec_yds
FROM advstats a
JOIN draft_picks d ON a.pfr_id = d.pfr_player_id
LEFT JOIN combine c ON d.pfr_player_id = c.pfr_id
WHERE d.position_group IN ('WR','RB')
  AND a.stat_type = 'rec'
GROUP BY speed_tier, a.career_year
HAVING COUNT(DISTINCT a.pfr_id) >= 5
ORDER BY a.career_year, speed_tier;
""", conn)
display(speed_traj)
```

Receiving yards by career year and combine speed tier:

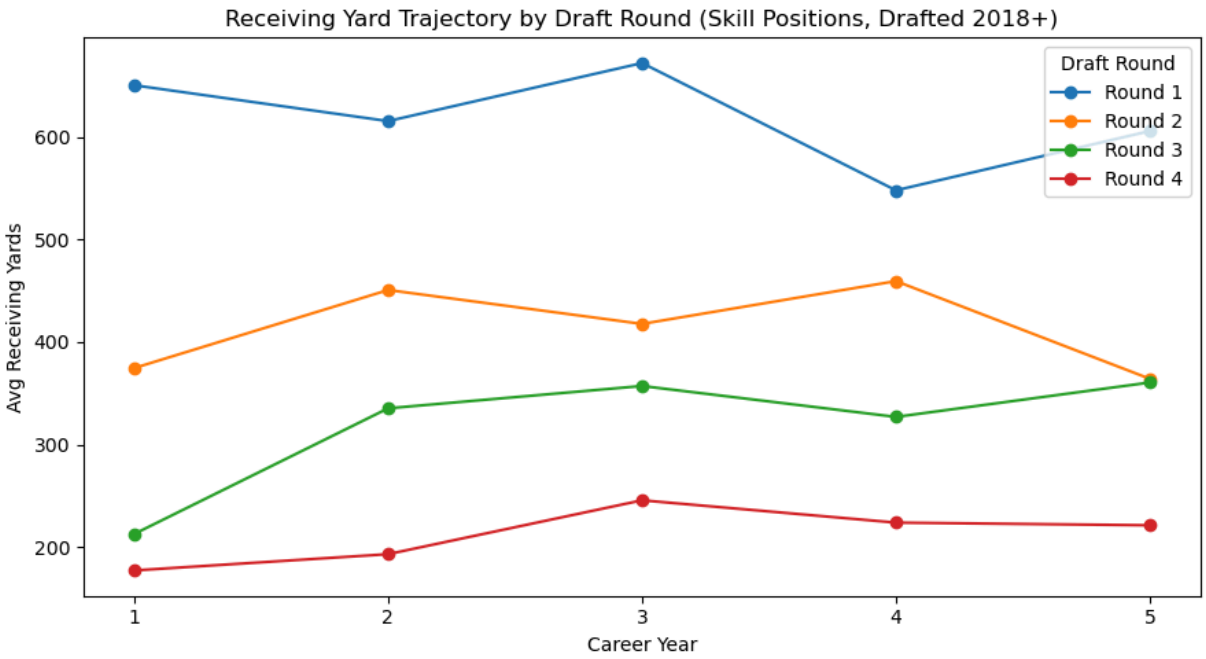
	speed_tier	career_year	players	avg_rec_yds
0	Fast (< 4.50s)	1.0	158	325.0
1	No forty recorded	1.0	90	239.4
2	Slower (4.50s+)	1.0	98	196.1
3	Fast (< 4.50s)	2.0	142	369.1
4	No forty recorded	2.0	82	303.4
5	Slower (4.50s+)	2.0	106	285.0
6	Fast (< 4.50s)	3.0	122	396.3
7	No forty recorded	3.0	61	392.8
8	Slower (4.50s+)	3.0	103	305.9
9	Fast (< 4.50s)	4.0	110	357.9
10	No forty recorded	4.0	48	280.8
11	Slower (4.50s+)	4.0	100	302.6
12	Fast (< 4.50s)	5.0	85	441.5
13	No forty recorded	5.0	36	313.1
14	Slower (4.50s+)	5.0	89	310.2

Visualization receiving yard trajectories by draft round

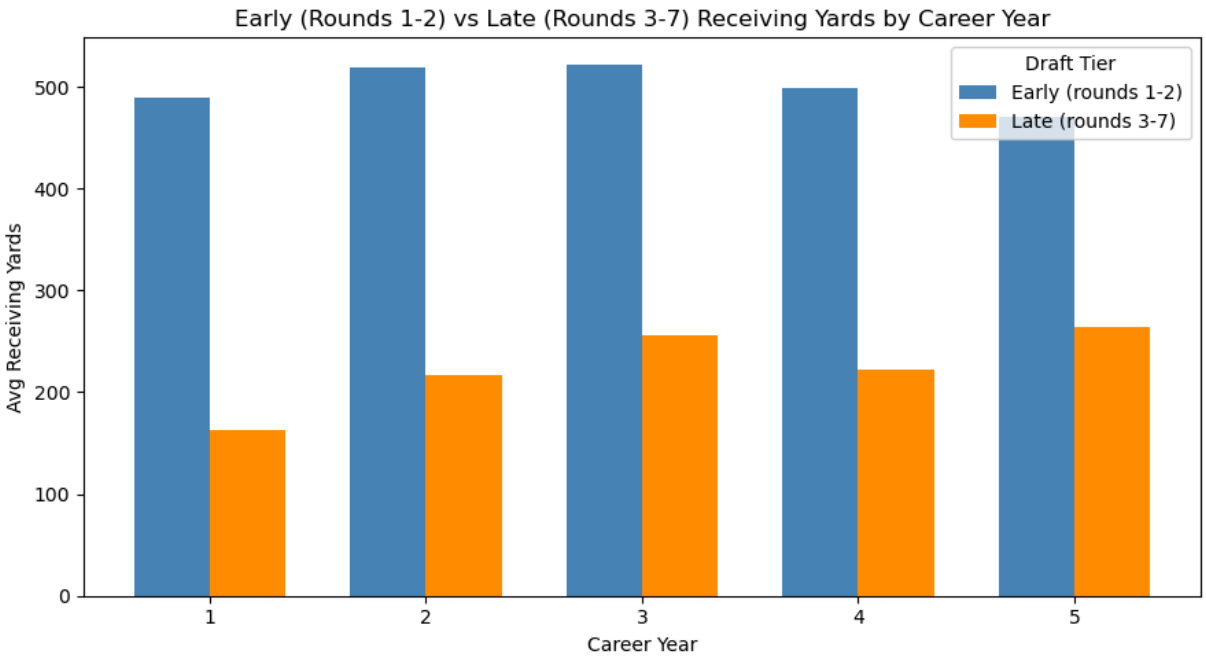
In [46]:
```
# Line chart showing average receiving yards per career year by draft round
fig, ax = plt.subplots(figsize=(9, 5))

for rnd in sorted(traj_rec['round'].unique()):
    subset = traj_rec[traj_rec['round'] == rnd]
    ax.plot(subset['career_year'], subset['avg_rec_yds'], marker='o', label=

ax.set_xlabel('Career Year')
ax.set_ylabel('Avg Receiving Yards')
ax.set_title('Receiving Yard Trajectory by Draft Round (Skill Positions, Dra
ax.legend(title='Draft Round')
ax.set_xticks([1, 2, 3, 4, 5])
plt.tight_layout()
plt.show()
```



In [47]:
```
# Bar chart comparing early vs late round picks across career years
early = tier_compare[tier_compare['draft_tier'] == 'Early (rounds 1-2)'].set
late  = tier_compare[tier_compare['draft_tier'] == 'Late (rounds 3-7)'].set_
x = range(len(early))
width = 0.35
fig, ax = plt.subplots(figsize=(9, 5))
ax.bar([i - width/2 for i in x], early.values, width, label='Early (rounds 1
ax.bar([i + width/2 for i in x], late.values,  width, label='Late (rounds 3-
ax.set_xticks(list(x))
ax.set_xticklabels(early.index.astype(int))
ax.set_xlabel('Career Year')
ax.set_ylabel('Avg Receiving Yards')
ax.set_title('Early (Rounds 1-2) vs Late (Rounds 3-7) Receiving Yards by Car
ax.legend(title='Draft Tier')
plt.tight_layout()
plt.show()
```



Section 8 - Findings

Conclusions

- **Finding 1**: Round 1 skill players averaged 650.7 receiving yards in career year 1, compared to 212.6 for round 3 players and 177.0 for round 4 players, a gap of over 470 yards in the first season. By career year 3 the round 1 average was 672.4 yards versus 357.1 for round 3, meaning the gap narrowed from 438 yards to 315 yards over two seasons. Even though this supports the hypothesis that later round picks close the gap over time among those who remain in the league, I though the gap would be closed more than 123 yeards.
- **Finding 2**: Early round picks (rounds 1-2) averaged significantly more receiving yards than late round picks (rounds 3-7) in every career year, but the gap narrowed from year 1 to year 3. Round 2 players started at 374.6 yards in year 1 and grew to 450.8 in year 2 and 417.7 in year 3, showing more consistent improvement than round 1 players who actually dipped from 650.7 in year 1 to 615.7 in year 2 before recovering to 672.4 in year 3.
- **Finding 3**: Players with sub 4.50 forty times produce more receiving yards in career year 1 than slower players at the same positions, but the advantage shrinks by career year 3. This suggests combine speed predicts early production more strongly than long term development trajectory.

Limitations

- The advstats data only goes back to 2018, meaning the sample of players with five full seasons of data is very small, most players in the dataset have only two or

three career years available. Findings about years 4 and 5 should be interpreted cautiously.
- Survival bias is a significant issue here. Late round picks who are still in the league by year 3 are not a random sample, they are the best performers from that draft tier. This makes the gap narrowing result look more optimistic than it may actually be for the average late round pick.

Follow-up questions

- Would the same trajectory patterns hold for a longer historical window if older advstats data becomes available?
- Does position matter within the skill group, do WRs develop differently than RBs across the same career year timeline?

In [49]:
```
conn.close()
```